COMPUTER ENGINEERING

QUEZON CITY

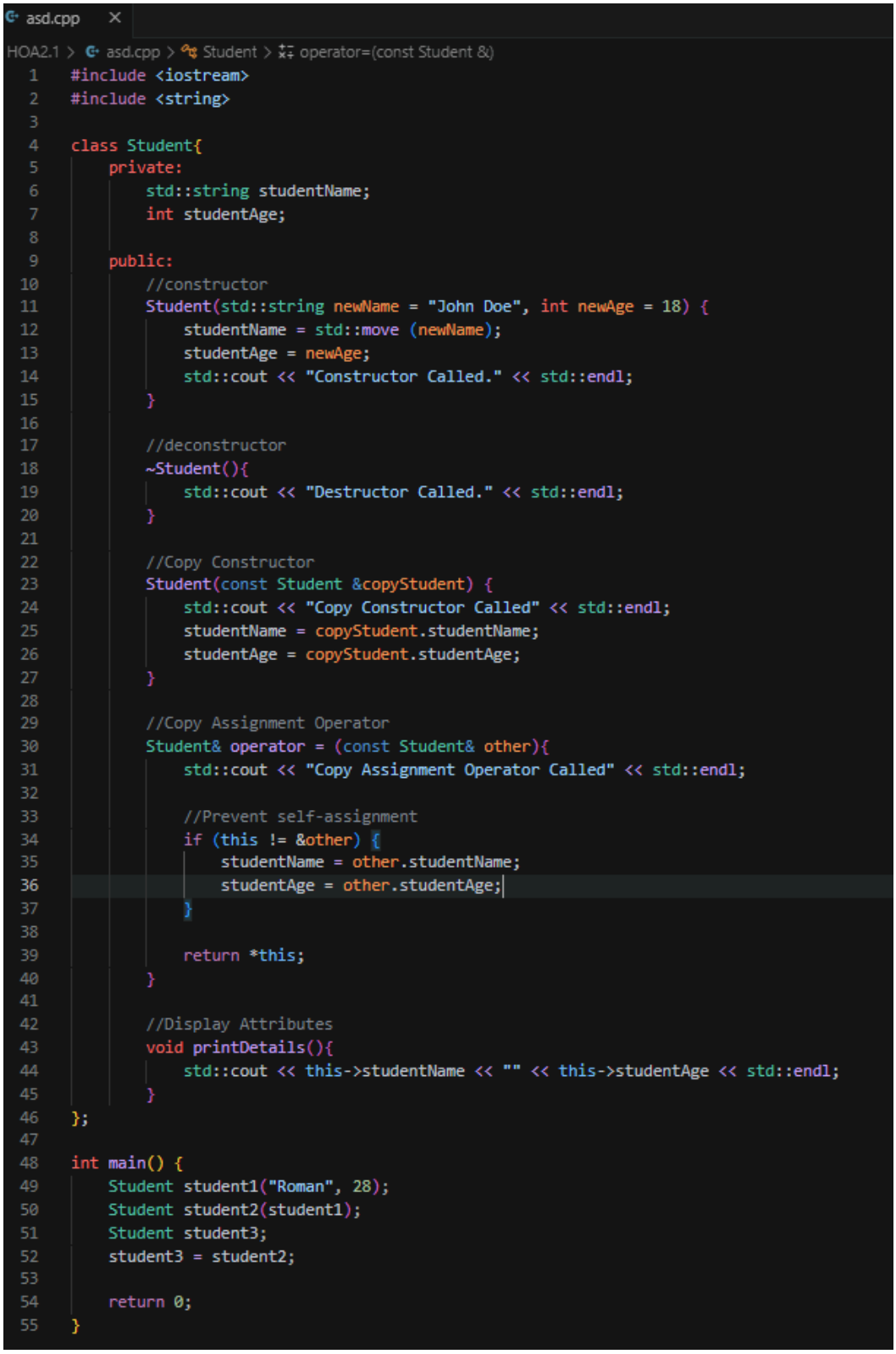
COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	07/17/2026	**DATE SUBMITTED:**	
NAME:	Kherwin Emmanuel D. Milan	**INSTRUCTOR:**	Jimlord M. Quejado

Hands-on Activity 2.1 Arrays, Pointers and Dynamic Memory Allocation

1. PROCEDURE OUTPUT

What procedures did you perform?

Screenshot


asd.cpp
HOA2.1 > asd.cpp > Student > operator=(const Student &)
#include <iostream>
#include <string>
class Student{
private:
std::string studentName;
int studentAge;
public:
//constructor
Student(std::string newName = "John Doe", int newAge = 18) {
studentName = std::move (newName);
studentAge = newAge;
std::cout << "Constructor Called." << std::endl;
}
//deconstructor
~Student(){
std::cout << "Destructor Called." << std::endl;
}
//Copy Constructor
Student(const Student &copyStudent) {
std::cout << "Copy Constructor Called" << std::endl;
studentName = copyStudent.studentName;
studentAge = copyStudent.studentAge;
}
//Copy Assignment Operator
Student& operator = (const Student& other){
std::cout << "Copy Assignment Operator Called" << std::endl;
//Prevent self-assignment
if (this != &other) {
studentName = other.studentName;
studentAge = other.studentAge;
}
return *this;
}
//Display Attributes
void printDetails(){
std::cout << this->studentName << "" << this->studentAge << std::endl;
}
};
int main() {
Student student1("Roman", 28);
Student student2(student1);
Student student3;
student3 = student2;
return 0;
}

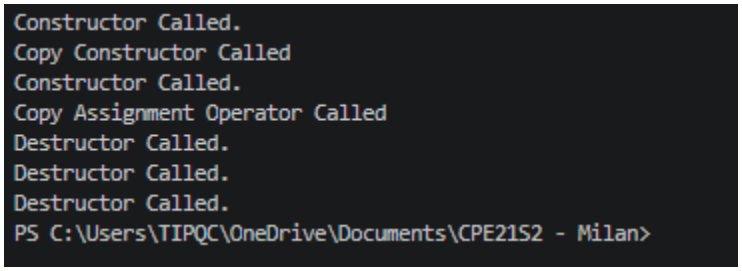
	
Observation	I observed how the Student objects were created and copied step by step. The constructor was called when student1 and student3 were created, while the copy constructor was called when student2 was created using student1. The copy assignment operator was called when the values of student2 were assigned to the already existing student3.

Table 2-1. Initial Driver Program

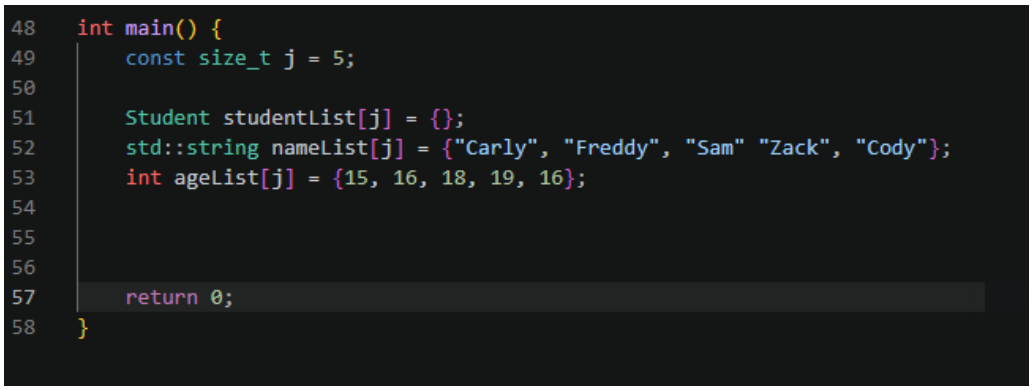
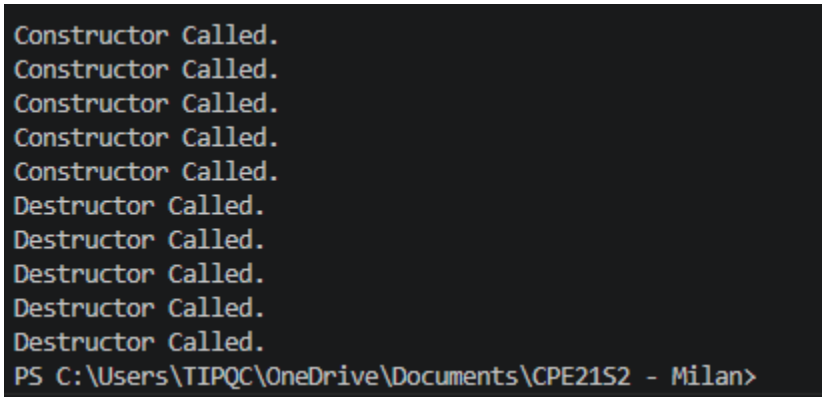
Screenshot	 
Observation	

Table 2-2. Modified Driver Program with Student Lists

Loop A	
Observation	
Loop B	
Observation	
Output	
Observation	

Table 2-3. Final Driver Program

Modifications	
Observation	

Table 2-4. Modifications/Corrections Necessary

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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